```
100 'FFT CONVOLUTION
110 'This program convolves a 10 million point signal with a 400 point filter kernel.  The input
120 'signal is broken into 16000 segments, each with 625 points.  1024 point FFTs are used.
130 '
130 '                                   'INITIALIZE THE ARRAYS
140 DIM XX[1023]                        'the time domain signal (for the FFT)
150 DIM REX[512]                        'real part of the frequency domain (for the FFT)
160 DIM IMX[512]                        'imaginary part of the frequency domain (for the FFT)
170 DIM REFR[512]                       'real part of the filter's frequency response
180 DIM IMFR[512]                       'imaginary part of the filter's frequency response
190 DIM OLAP[398]                       'holds the overlapping samples from segment to segment
200 '
210 FOR I% = 0 TO 398                   'zero the array holding the overlapping samples
220   OLAP[I%] = 0
230 NEXT I%
240 '
250 '                                   'FIND & STORE THE FILTER'S FREQUENCY RESPONSE
260 GOSUB XXXX                          'Mythical subroutine to load the filter kernel into XX[ ]
270 GOSUB XXXX                          'Mythical FFT subroutine:  XX[ ] -->  REX[ ] & IMX[ ]
280 FOR F% = 0 TO 512                   'Save the frequency response in REFR[ ] & IMFR[ ]
290   REFR[F%] = REX[F%]
300   IMFR[F%] = IMX[F%]
310 NEXT F%
320 '
330 '                                   'PROCESS EACH OF THE 16000 SEGMENTS
340 FOR SEGMENT% = 0 TO 15999
350   '
360   GOSUB XXXX                        'Mythical subroutine to load the next input segment into XX[ ]
370   GOSUB XXXX                        'Mythical FFT subroutine:  XX[ ]  -->  REX[ ] & IMX[ ]
380   '
390   FOR F% = 0 TO 512   'Multiply the frequency spectrum by the frequency response
400     TEMP      = REX[F%]*REFR[F%] - IMX[F%]*IMFR[F%]
410     IMX[F%]   = REX[F%]*IMFR[F%] + IMX[F%]*REFR[F%]
420     REX[F%]   = TEMP
430   NEXT F%
440   '
450   GOSUB XXXX                        'Mythical IFFT subroutine:  REX[ ] & IMX[ ]  -->  XX[ ]
460   '
470   FOR I% = 0 TO 398    'Add the last segment's overlap to this segment
480     XX[I%] = XX[I%] + OLAP[I%]
490   NEXT I%
500   '
510   FOR I% = 625 TO 1023              'Save the samples that will overlap the next segment
520     OLAP[I%-625] = XX[I%]
530   NEXT I%
540   '
550   GOSUB XXXX                        'Mythical subroutine to output the 625 samples stored
560   '                                 'in XX[0] to XX[624]
570   '
580 NEXT SEGMENT%
590 '
600 GOSUB XXXX                          'Mythical subroutine to output all 399 samples in OLAP[ ]
610 END
```

TABLE 18-1

for standard convolution is directly proportional to the number of points in the filter kernel. In comparison, the time required for FFT convolution increases very slowly, only as the *logarithm* of the number of points in the